

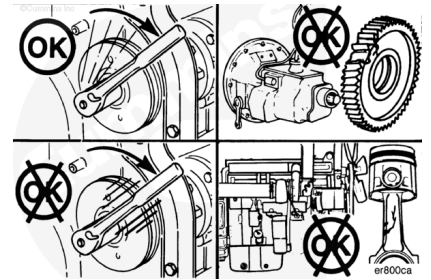
Trainings ▾

Rotation Check

Use **only** the accessory driveshaft to rotate the crankshaft.

Rotate the crankshaft **clockwise** through two complete revolutions.

If the engine does **not** turn freely, the equipment can have a malfunction. Refer to the OEM service manual. The engine can have internal problems. Reference the appropriate procedure for inspection and replacement of internal engine components.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the engine. Refer to Procedure 000-001 in Section 0. (</qs3/pubsys2/xml/en/procedures/35/35-000-001-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html>)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-006-tr.html>)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Remove the front gear housing. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-033-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Remove the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-040-tr.html>)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-031-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-089-tr.html)

- Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html)

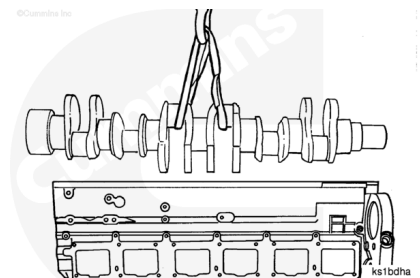
- Remove the main bearing caps. Refer to Procedure 001-006 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-006-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a lifting strap that will **not** damage the crankshaft.

Use a hoist and a lifting strap to remove the crankshaft.

Clean and Inspect for Reuse

Use a crocus cloth, or a Scotch-Brite™ pad to remove discoloration or light scratches from the machined surfaces of the crankshaft.

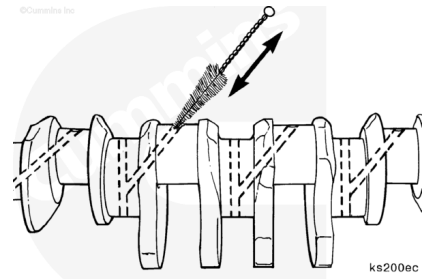


⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

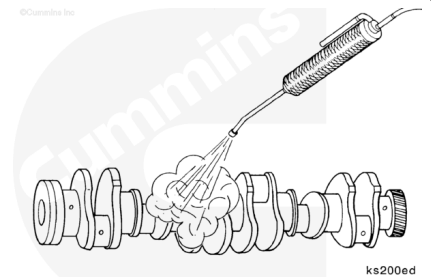
possibility of personal injury.

Use a bristle brush and solvent to clean all the crankshaft oil drillings.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



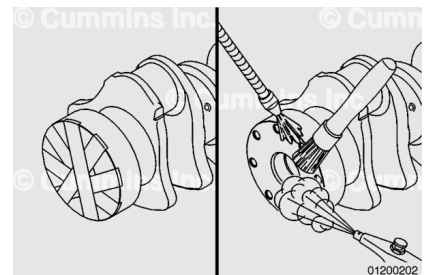
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the crankshaft and dry with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a thread chaser to clean the capscrew threads in the crankshaft, severe engine damage can result.

To clean the rolled threads, flush with solvent, and dry with compressed air.

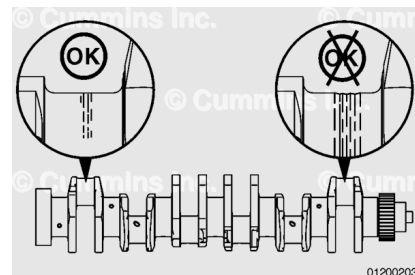
If additional cleaning is required, brush with a nylon bristle brush.

Place tape over the threaded capscrew holes.

Inspect the machined surfaces for scratches or nicks.

Use a crocus cloth to remove any nicks and scratches.

If scratches or nicks can be felt with a fingernail after the crankshaft has been polished with crocus cloth, the crankshaft **must** be replaced.



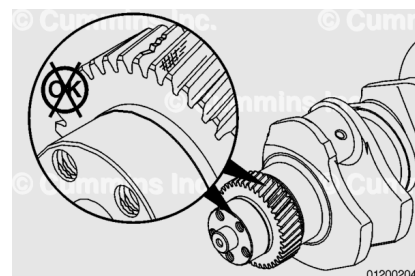
Inspect the threaded capscrew holes for damage.

If the threaded capscrew holes are damaged, the crankshaft **must** be replaced.

Inspect the crankshaft gear for excessive wear or damage.

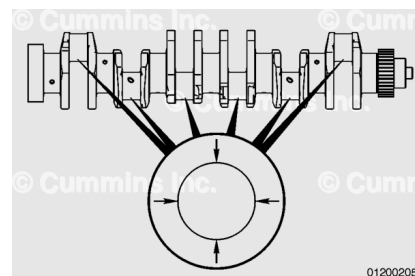
Note : If no damage exists, do not remove the gear.

If any teeth on the crankshaft gear are missing, chipped, or pitted, the crankshaft gear **must** be replaced.



Measure the crankshaft connecting rod journal outside diameter.

Crankshaft Connecting Rod Journal Outside Diameter		
mm		in
78.987	MIN	3.1097
79.013	MAX	3.1107



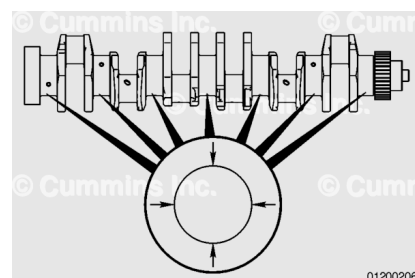
If the crankshaft connecting rod journal outside diameter is **not** within specifications, the crankshaft **must** be replaced.

Note : Bearings are available for both connecting rod and main bearings journals to accommodate 0.25 mm [0.010 in], 0.50 mm [0.020 in], 0.75 mm [0.030 in] and 1.00 mm [0.039 in] undersize bearings journals. Reference the Alternative Repair Manual, L10 and M11 Series Engines for further information.

Measure the crankshaft main bearing journal outside diameter.

Crankshaft Main Bearing Journal Outside Diameter

mm		in
114.015	MIN	4.4888
114.055	MAX	4.4904



If the crankshaft main bearing journal outside diameter is **not** within specifications, the crankshaft **must** be replaced.

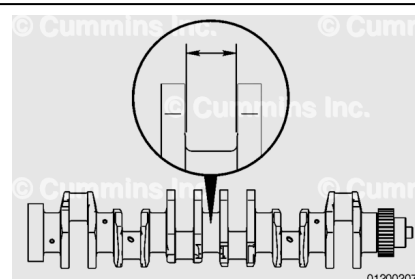
Note : Bearings are available for both connecting rod and main bearings journals to accommodate 0.25 mm [0.010 in], 0.50 mm [0.020 in], 0.75 mm [0.030 in] and 1.00 mm [0.039 in] undersize bearings journals.

Reference the Alternative Repair Manual, L10 and M11 Series Engines for further information.

Measure the crankshaft thrust face width.

Crankshaft Thrust Face Width

mm		in
49.975	MIN	1.9675
50.100	MAX	1.9724



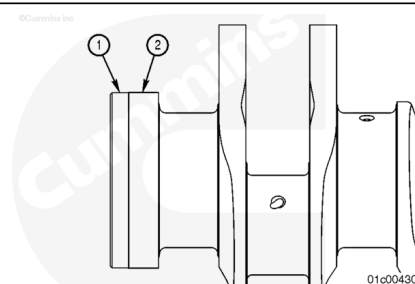
If the crankshaft thrust face width is **not** within specifications, the crankshaft **must** be replaced.

Note : There are 0.25 mm [0.010 in] oversize thrust bearings available. Reference the Alternative Repair Manual, L10 and M11 Series Engines for further information.

Measure the crankshaft rear oil seal flange outside diameter.

Crankshaft Rear Oil Seal Flange Outside Diameter

mm		in
164.965	MIN	6.4957
165.035	MAX	6.4974



If the crankshaft rear oil seal flange outside diameter is **not** within specifications, the crankshaft **must** be replaced.

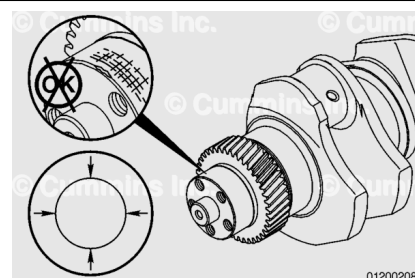
Check the crankshaft rear oil seal flange for nicks, burrs, and grooves.

If a fingernail catches in a nick, burr, or groove, the crankshaft **must** be replaced. Damage to the sealing surface can result in a seal leak.

Note : If the rear oil seal flange requires repair, a wear sleeve can be installed. Installation of a wear sleeve usually does **not** require machining of the crankshaft. Reference the Alternative Repair Manual, L10 and M11 Series Engines for further information.

Measure the crankshaft front oil seal flange outside diameter.

Crankshaft Front Oil Seal Flange Outside Diameter		
mm		in
84.970	MIN	3.3453
85.000	MAX	3.3465



If the crankshaft front oil seal flange outside diameter is **not** within specifications, the crankshaft **must** be replaced.

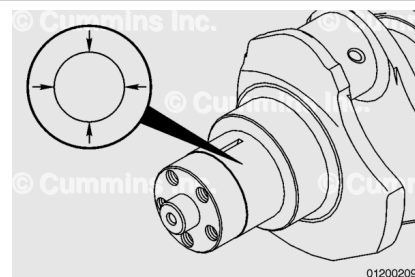
Check the crankshaft front oil seal flange for nicks, burrs, and grooves.

If a fingernail catches in a nick, burr, or groove, the crankshaft **must** be replaced. Damage to the sealing surface can result in a seal leak.

Note : If the front oil seal flange requires repair, a wear sleeve can be installed. Installation of a wear sleeve usually does **not** require machining of the crankshaft. Reference the Alternative Repair Manual, L10 and M11 Series Engines for further information.

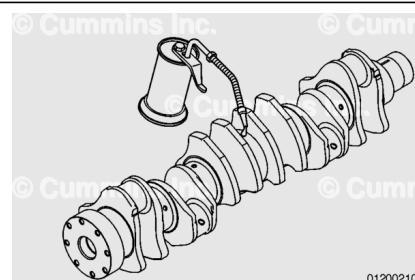
Measure the crankshaft gear journal diameter, if the crankshaft gear has been removed.

Crankshaft Gear Journal Diameter		
mm		in
85.975	MIN	3.3848
86.000	MAX	3.3858



If the crankshaft gear journal diameter is **not** within specifications, the crankshaft **must** be replaced.

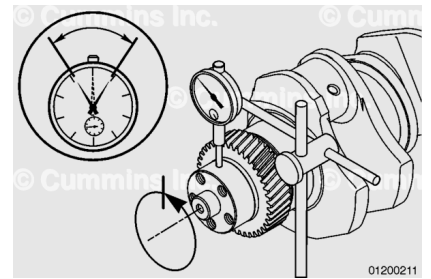
Lubricate the entire crankshaft with clean 15W-40 oil to prevent rust.



Bend and Twist Inspect

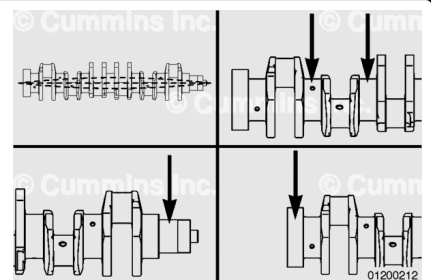
The crankshaft alignment limits are based on total indicator runout (TIR).

The runout is measured as the crankshaft is turned one revolution, recording the total movement of the dial indicator.



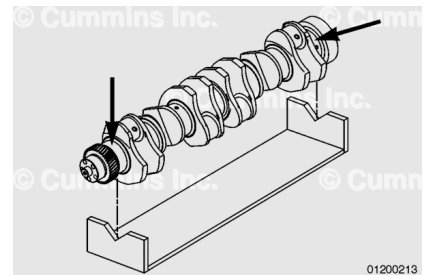
When measuring the crankshaft alignment, four measurements **must** be taken:

- Full length alignment
- Bearing to bearing alignment
- Gear journal diameter runout
- Rear oil seal flange diameter runout.



⚠ WARNING ⚠

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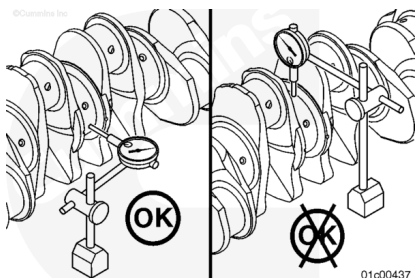


⚠ CAUTION ⚠

Measuring from the top of the main bearing journal will cause inaccurate alignment measurements because of crankshaft sag.

To check the crankshaft full length alignment, support the crankshaft on the number one and number seven main bearing journals.

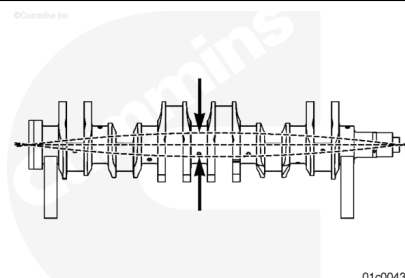
Set up the dial indicator. The dial indicator **must** be set up so the indicator is located on the side of the number four main bearing journal.



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Rotate the crankshaft one complete revolution and record the total indicator runout (TIR).

Total Indicator Runout (TIR)		
mm		in
0.15	MAX	0.006



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If the crankshaft total indicator runout (TIR) is **not** within specifications, the crankshaft **must** be replaced.

To check the crankshaft bearing to bearing alignment, support the crankshaft on the number one and number seven main bearing journals.

Note : The dial indicator must be set up so the indicator is located on the side of the bearing journal being measured.

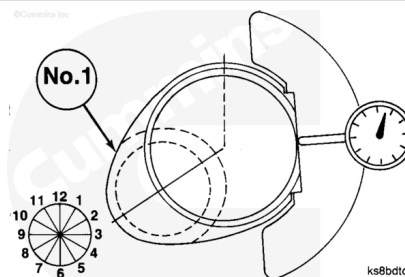
Rotate the crankshaft one complete revolution and record the total indicator runout (TIR) for each main bearing journal. The total indicator runout (TIR) of each journal must be within 0.05 mm [0.002 in] of the adjacent journal.

If the crankshaft total indicator runout (TIR) is **not** within specifications, the crankshaft **must** be replaced.

Crankshaft gear journal diameter runout measurement:

With the crankshaft supported on the number 1 and the number 7 main bearing journals, install a dial indicator within 6.4 mm [0.252 in] from the front end of the gear journal.

Rotate the crankshaft one complete revolution and record the total indicator runout (TIR).



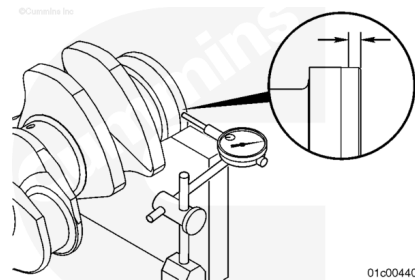
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Crankshaft Gear Journal Diameter Total Indicator Runout (TIR)

mm		in
0.030	MAX	0.0012

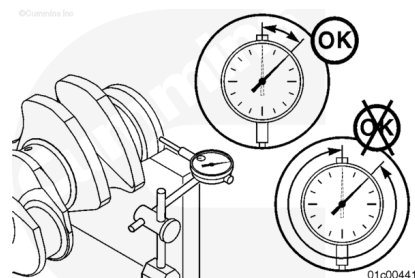
Rear oil seal flange diameter runout measurement:

With the crankshaft supported on the number one and number seven main bearing journals, position a dial indicator within 6.4 mm [0.252 in] from the rear of the oil seal flange diameter.



Rotate the crankshaft one complete revolution and record the total indicator runout (TIR).

Rear Oil Seal Flange Total Indicator Runout (TIR)		
mm		in
0.030	MAX	0.0012



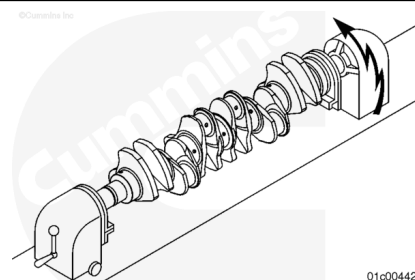
If the rear oil seal flange total indicator runout (TIR) is **not** within specifications, the crankshaft **must** be replaced.

Magnetic Crack Inspect

The crankshaft **must** be tested by the “continuous method”. The entire surface **must** be wetted with the magnetic bath solution before and during the flow of magnetic current through the crankshaft.

For the head shot (longitudinal magnetization) method, apply the magnetic bath solution to the crankshaft and use 3800 amperes with VDC or rectified VAC to magnetize the crankshaft.

Note : The head shot method passes the magnetizing current longitudinally through the crankshaft. The head shot produces a circular magnetic field which is most effective in detecting longitudinal defects in the crankshaft.



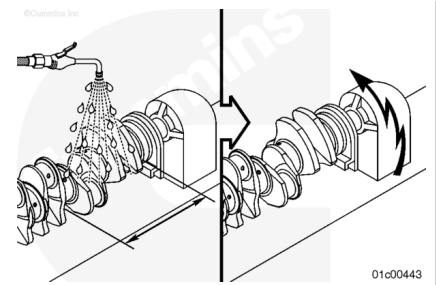
A minimum of three successive head shots are required for each section of the crankshaft.

Flow the magnetic solution over the first 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply a third head shot of magnetizing current.

Inspect the crankshaft for indications.

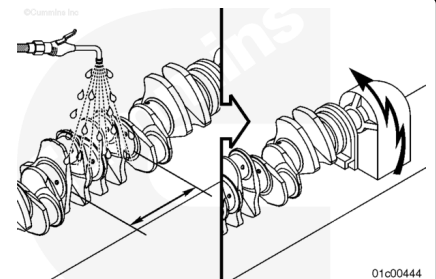


Flow the magnetic solution over the second 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply a third head shot of magnetizing current.

Inspect the crankshaft for indications.

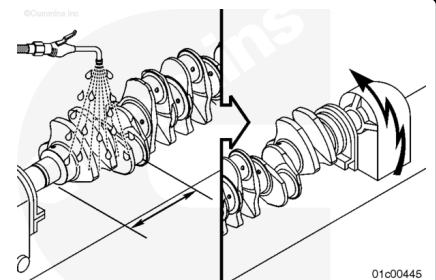


Flow the magnetic solution over the last 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply a third shot of magnetizing current.

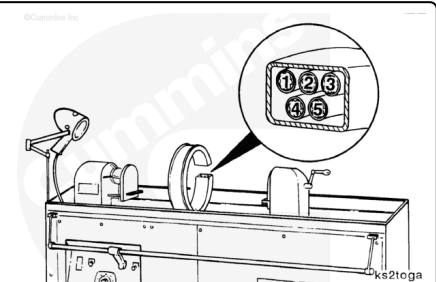
Inspect the crankshaft for indications.



For the coil shot (circumferential magnetization) method, apply the magnetic solution and magnetize the crankshaft in a 514.5 mm [20.256 in] inside coil or equivalent. Use 3,600 to 4,000 ampere-turns with VDC or rectified VAC.

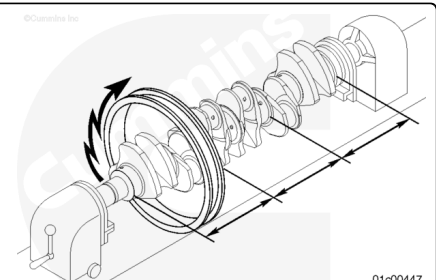
Note : The coil shot method passes the magnetizing current through a coil centered around the crankshaft axis. The coil shot produces a longitudinal magnetic field which is most effective in detecting circumferential defects.

Ampere-turn is the amperage flowing through the coil multiplied by the number of turns in the coil.



A minimum of three coil shots are required to correctly magnetize the crankshaft.

Flow the magnetic solution over the crankshaft in 1/3 sections.



Apply two coil shots, after wetting with the magnetic solution, with the coil placed at ¼-length points along the crankshaft.

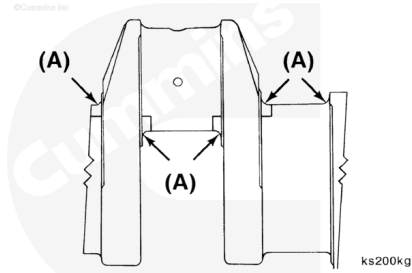
To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply the third coil shot with the coil placed at ¼-length points along the crankshaft.

Inspect each 457 mm [18 in] section after the third coil shot is completed.

Do **not** mistake forging trim lines or flow lines for crack indications.

Acceptability of indications varies depending on their location on the crankshaft. Reference the table below.



	Open Indication	Subsurface Indication	Circumferen tial Indication
Fillets	Indications larger than 1.50 mm [0.060 in] are not acceptable in fillets or the critical areas (see note).	Indications larger than 1.50 mm [0.060 in] are not acceptable in fillets or the critical areas (see note).	

	Open Indication	Subsurface Indication	Circumferential Indication
Connecting Rod Bearing Journals	Indications in the longitudinal direction beyond the individual maximum length or the total cumulative length of open longitudinal indications on any one connecting rod journal as indicated below are not acceptable. Total cumulative length is 44.50 mm [1.752 in] max. Individual length is 12.75 mm [0.502 in] max.	Subsurface circumferential indications are not acceptable.	Indications are not acceptable.

	Open Indication	Subsurface Indication	Circumferential Indication
Main Bearing Journals	Indications in the longitudinal direction beyond the individual maximum length or the total cumulative length of open longitudinal indications on any one main bearing journal as indicated below are not acceptable. Total cumulative length is 63.50 mm [2.500 in] max. Individual length is 15.896 mm [0.626 in] max.	Subsurface indications no longer than 28.50 mm [1.122 in] are acceptable.	Indications are not acceptable.
Bearing Fillet Sidewalls	Longitudinal indications no longer 4.750 mm [0.1870 in] in or entering the fillet are acceptable.	Longitudinal indications no longer than 6.50 mm [0.256 in] are acceptable.	Indications of any kind are not acceptable.

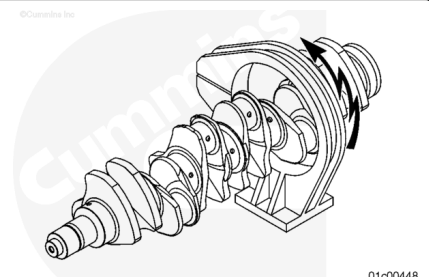
	Open Indication	Subsurface Indication	Circumferential Indication
Oil Drillings	Indications entering the oil hole or chamfer are not acceptable. Indications in the longitudinal direction up to 12.50 mm [0.492 in] long but not closer than 3 mm [0.118 in] to the oil hole chamfer are acceptable. Indication in the longitudinal direction closer than 3 mm [0.118 in] to the oil hole chamfer are not acceptable.	Indications entering the oil hole or chamfer are not acceptable. Indications in the longitudinal direction not closer than 3 mm [0.118 in] to the oil hole chamfer and not more than 12.50 mm [0.492 in] long are acceptable. Indications in the longitudinal direction closer than 3 mm [0.118 in] to the oil hole chamfer.	Indications are not acceptable.

	Open Indication	Subsurface Indication	Circumferential Indication
Crankshaft Webs	Indications in the critical area are not acceptable (see note). Indications that extend into the rod journal fillet are not acceptable. Indications are not acceptable.	Indications in the critical area are not acceptable (see note). Indications that extend into the rod journal fillet are not acceptable. Indications that extend out to the cast surface are acceptable.	Indications that extend into the rod journal fillet are not acceptable.
Gear Journal Diameter	Indications that enter the crankshaft keyway are not acceptable. Indications longer than 1.50 mm [0.060 in] are not acceptable. Indication that extends over the face of the crankshaft or into the damper mounting capscrew holes is not acceptable.	Indications in the longitudinal direction no longer than 50 mm [1.97 in] are acceptable. Indication that extends over the face of the crankshaft or into the damper mounting capscrew holes is not acceptable.	Indications are not acceptable.

	Open Indication	Subsurface Indication	Circumferential Indication
Rear Oil Seal Flange Diameter	Indications are not acceptable.	Indications no longer than 1.50 mm [0.060 in] are acceptable as long as the indication does not extend across the oil seal diameter chamfer.	Indications are not acceptable.
NOTE: The critical area extends from the fillet side wall tangency to 3 mm [0.118 in] beyond the fillet journal tangency on both the connecting rod and main bearing journal fillets.			

⚠ CAUTION ⚠

All magnetism must be removed from the crankshaft completely and the crankshaft cleaned thoroughly. Small metal particles will cause engine damage.



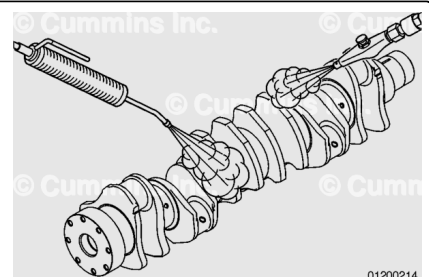
Demagnetize the crankshaft.

The maximum residual magnetic field on any pin or main bearing journal is 8 gauss.

It will possibly be necessary to use reverse D.C.-30 point step down equipment, or equivalent, to remove the magnetism.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.



⚠ WARNING ⚠

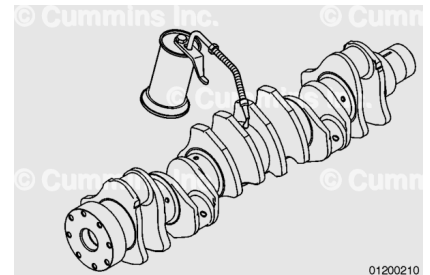
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the crankshaft.

Dry the crankshaft with compressed air.

Use a light preservative oil to lubricate the crankshaft to prevent rust.

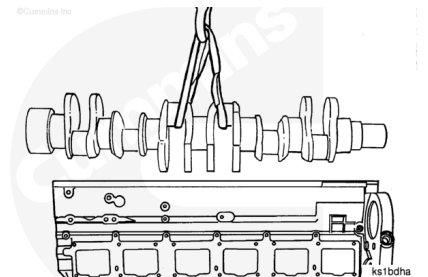
If the crankshaft is **not** going to be used immediately, protect it with a plastic cover to prevent dirt sticking to the oil.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Use a lifting strap that will not damage the crankshaft. Do not drop the crankshaft on the bearings.

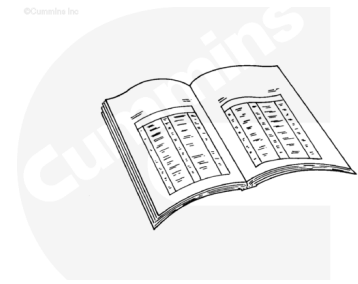
Install the main bearings. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-006-tr.html>)

The end of the crankshaft with the smallest diameter **must** point toward the front of the cylinder block.

Install the crankshaft.

Finishing Steps

- Install the main bearing caps. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-006-tr.html>)
- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html>)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-089-tr.html>)
- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-031-tr.html>)
- Install the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-040-tr.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Install the front gear housing. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-033-tr.html>)
- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-006-tr.html>)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)
- Install the engine. Refer to Procedure 000-002 in Section 0. (</qs3/pubsys2/xml/en/procedures/35/35-000-002-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.



(/qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine and check for leaks.

Last Modified: 28-Jul-2009
